

Lines and Planes in Three Dimensions



The vector equation of a line

- 1 Write the vector equation of the line through the point $(2, -1, 3)$ with direction vector $(1, 2, -1)$.
 - 2 Find the vector equation of the line through the two points $A(1, 0, 2)$ and $B(4, 3, 1)$.
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Determining if two lines intersect

- 3 Determine whether the lines $\mathbf{r}_1 = (1, 2, 0) + s(1, -1, 2)$ and $\mathbf{r}_2 = (0, 3, 1) + t(2, 1, -1)$ intersect, and if so, find the point of intersection.
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The equation of a plane

- 4 Find the Cartesian equation of the plane with normal vector $(2, -1, 3)$ passing through the point $(1, 2, -1)$.
 - 5 Find the Cartesian equation of the plane containing the points $A(1, 0, 0)$, $B(0, 1, 0)$, and $C(0, 0, 1)$.
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Angle between a line and a plane

- 6 Find the angle between the line with direction $(1, 2, 2)$ and the plane $x + y + z = 5$, to the nearest degree.
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Distance from a point to a plane

- 7 Find the shortest distance from the point $(3, 1, -2)$ to the plane $2x - y + 2z = 6$, using
$$d = \frac{|ax_0 + by_0 + cz_0 - k|}{\sqrt{a^2 + b^2 + c^2}}.$$
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Finding the angle between two planes

- 8 Find the angle between the planes $x + 2y - z = 3$ and $2x - y + z = 5$, to the nearest degree.
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Finding the intersection line of two planes

- 9 Find a vector equation for the line of intersection of the planes $x + y + z = 6$ and $x - y + 2z = 5$, using the cross product of the two normal vectors as the direction, and finding one point on both planes by inspection.
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Testing whether a line lies within a plane

- 10 Determine whether the line $\mathbf{r} = (1, 2, 3) + t(2, -1, 1)$ lies entirely within the plane $x + 3y + z = 10$.
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Finding the point where a line meets a plane

- 11 Find the point where the line $\mathbf{r} = (1, 0, 2) + t(1, 2, -1)$ intersects the plane $2x + y - z = 3$.
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Shortest distance between two skew lines

- 12 Two skew lines have direction vectors $\mathbf{d}_1 = (1, 0, 1)$ and $\mathbf{d}_2 = (0, 1, 1)$, and pass through the points $A(0, 0, 0)$ and $B(1, 1, 0)$ respectively. Find the shortest distance between the two lines, using $d = |\vec{AB} \cdot (\mathbf{d}_1 \times \mathbf{d}_2)| / |\mathbf{d}_1 \times \mathbf{d}_2|$.
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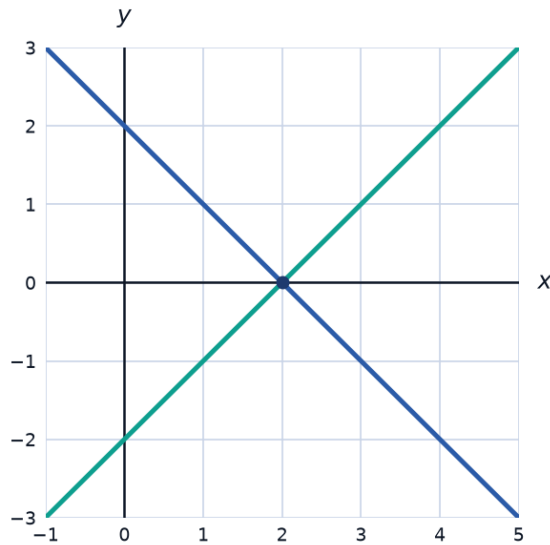
Foot of the perpendicular from a point to a line

- 13 Find the foot of the perpendicular from the point $P(4, 0, 1)$ to the line $\mathbf{r} = (1, 0, 0) + t(1, 1, 1)$.
- 14 Determine whether the planes $2x - y + 3z = 4$ and $-4x + 2y - 6z = -8$ represent the same plane, a pair of parallel (distinct) planes, or intersecting planes.
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Checking whether a point lies on a plane

- 15 Determine whether the point $P(2, -1, 4)$ lies on the plane $3x + 2y - z = 1$, and if not, state whether it lies on the same side as the origin.
- 16 Find the value of k such that the line $\mathbf{r} = (1, 2, 0) + t(2, -1, k)$ is parallel to the plane $x + 2y - z = 5$.
- 17 A surveyor models a hillside as the plane $2x + 3y + 6z = 42$, where x and y are horizontal coordinates (in metres) and z is height (in metres). Find the height z of the hillside at the point where $x = 6$ and $y = 4$, and find the plane's intercept on the z -axis (i.e. the height at $x = 0, y = 0$).

- 18 This question has three parts. Two lines are given by $\mathbf{r}_1 = (2, 0, 1) + s(1, 1, 0)$ and $\mathbf{r}_2 = (0, 2, 1) + t(1, -1, 0)$. (a) Show that both lines lie in the plane $z = 1$. (b) Find the point of intersection of the two lines. (c) Find the acute angle between the two lines, to the nearest degree.



Converting between vector and Cartesian forms of a line

- 19 Write the parametric equations of the line $\mathbf{r} = (2, 1, -3) + t(1, -2, 4)$, and hence give its Cartesian (symmetric) form.

Determining if two planes are parallel

- 20 Determine whether the planes $x - 2y + 3z = 5$ and $2x - 4y + 6z = 1$ are parallel, and if so, whether they represent the same plane.

Distance between two parallel planes

- 21 Find the distance between the parallel planes $x - 2y + 3z = 5$ and $2x - 4y + 6z = 1$ from the previous question.

Checking whether three points are collinear

- 22 Determine whether the points $A(1, 2, 3)$, $B(2, 4, 6)$, and $C(4, 8, 12)$ are collinear.
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Reflecting a point in a plane

23 Find the reflection of the point $P(1, 2, 0)$ in the plane $x + y + z = 6$.